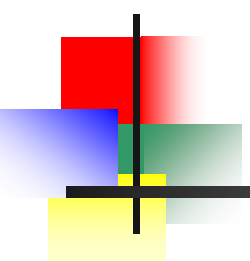


Learning Objectives

In this chapter, you learn:

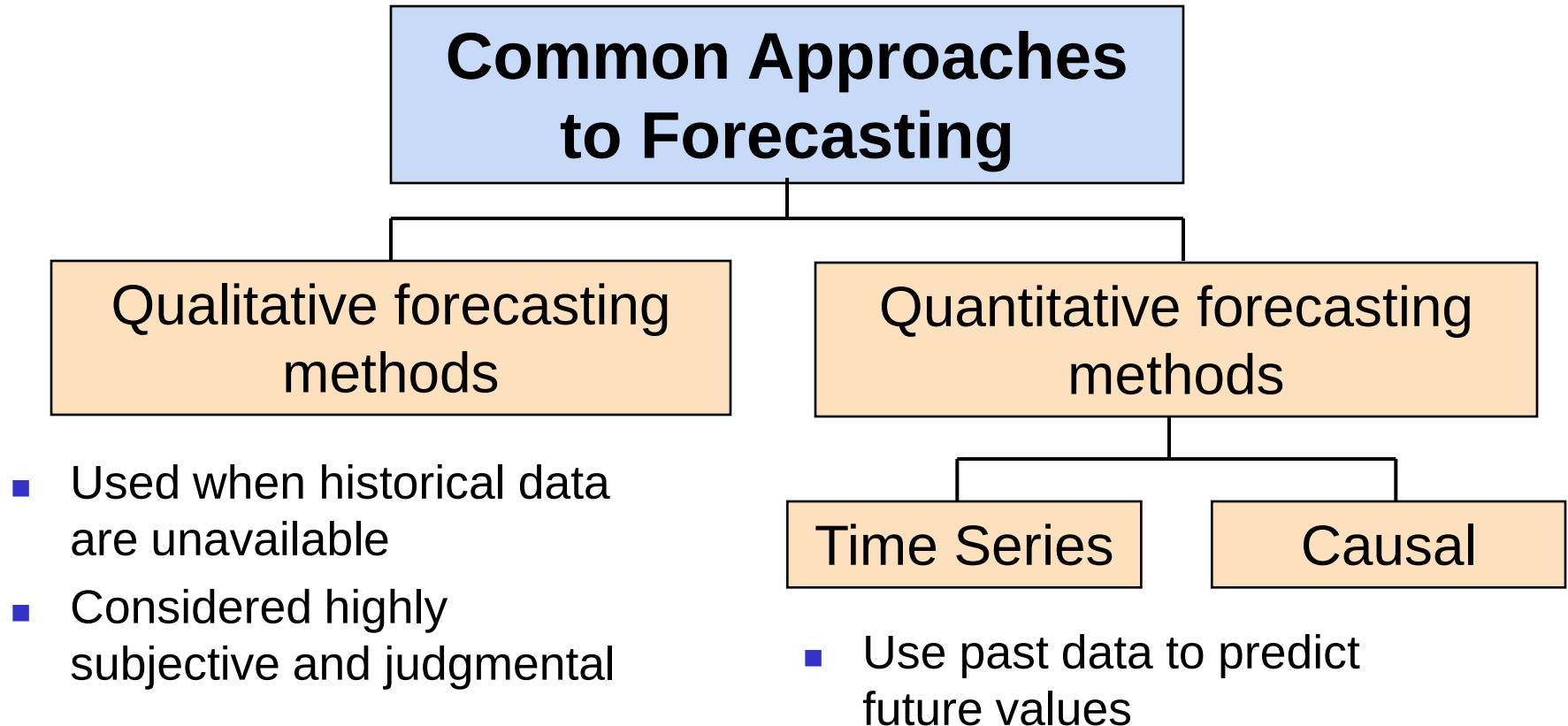
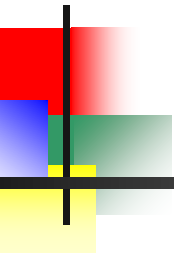
- How and when to use moving averages and exponential smoothing to smooth a time series
- To use linear trend, quadratic trend, and exponential trend time-series models
- To use the Holt-Winters forecasting model

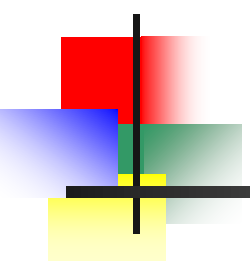


The Importance of Forecasting

- Governments forecast unemployment, interest rates, and expected revenues from income taxes for policy purposes
- Marketing executives forecast demand, sales, and consumer preferences for strategic planning
- College administrators forecast enrollments to plan for facilities and for faculty recruitment
- Retail stores forecast demand to control inventory levels, hire employees and provide training

Common Approaches to Forecasting





Time-Series Data

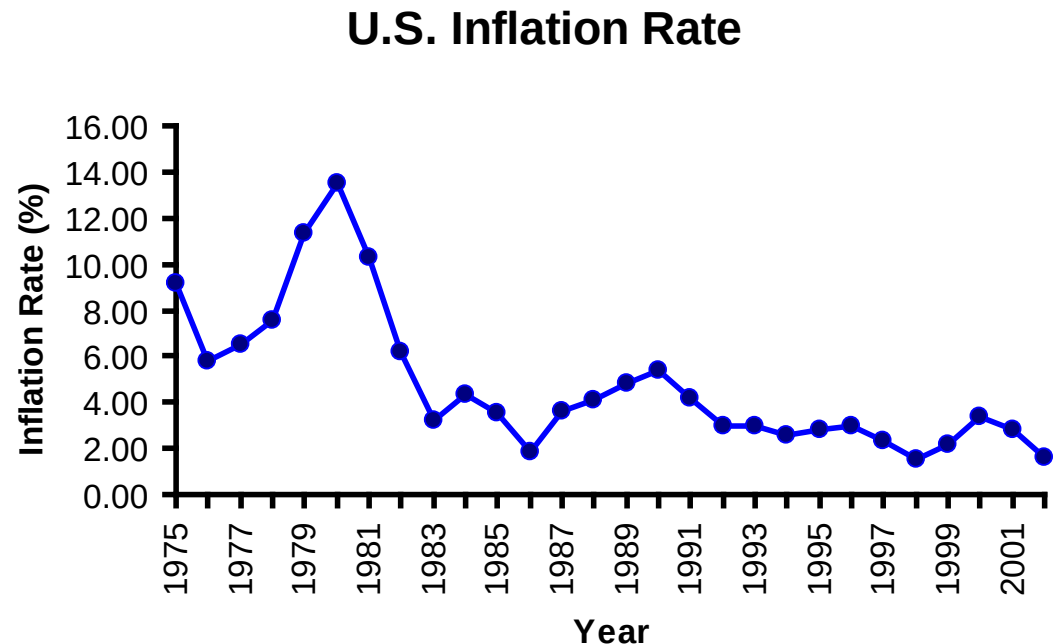
- Numerical data obtained at regular time intervals
- The time intervals can be annually, quarterly, daily, hourly, etc.
- Example:

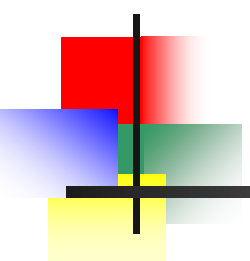
Year:	2000	2001	2002	2003	2004
Sales:	75.3	74.2	78.5	79.7	80.2

Time-Series Plot

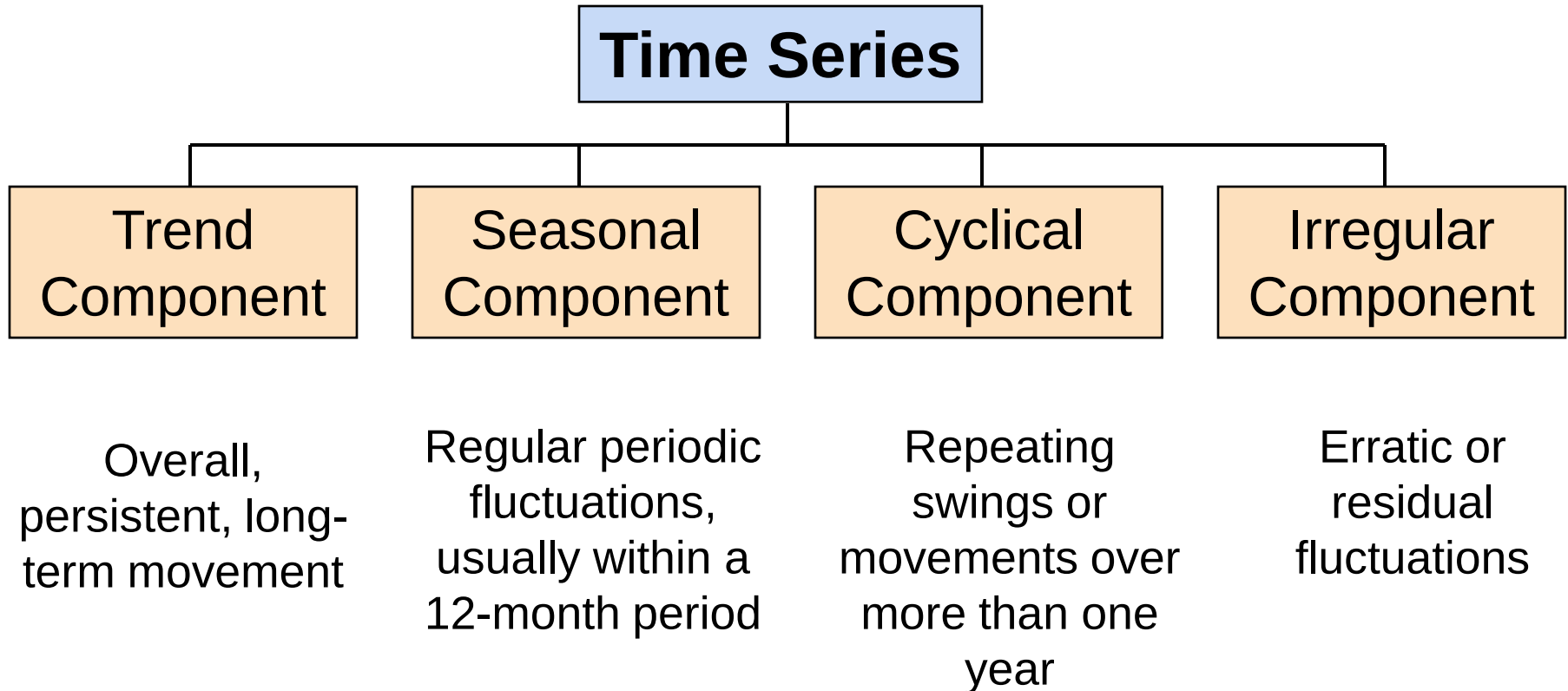
A **time-series plot** is a two-dimensional plot of time series data

- the vertical axis measures the variable of interest
- the horizontal axis corresponds to the time periods



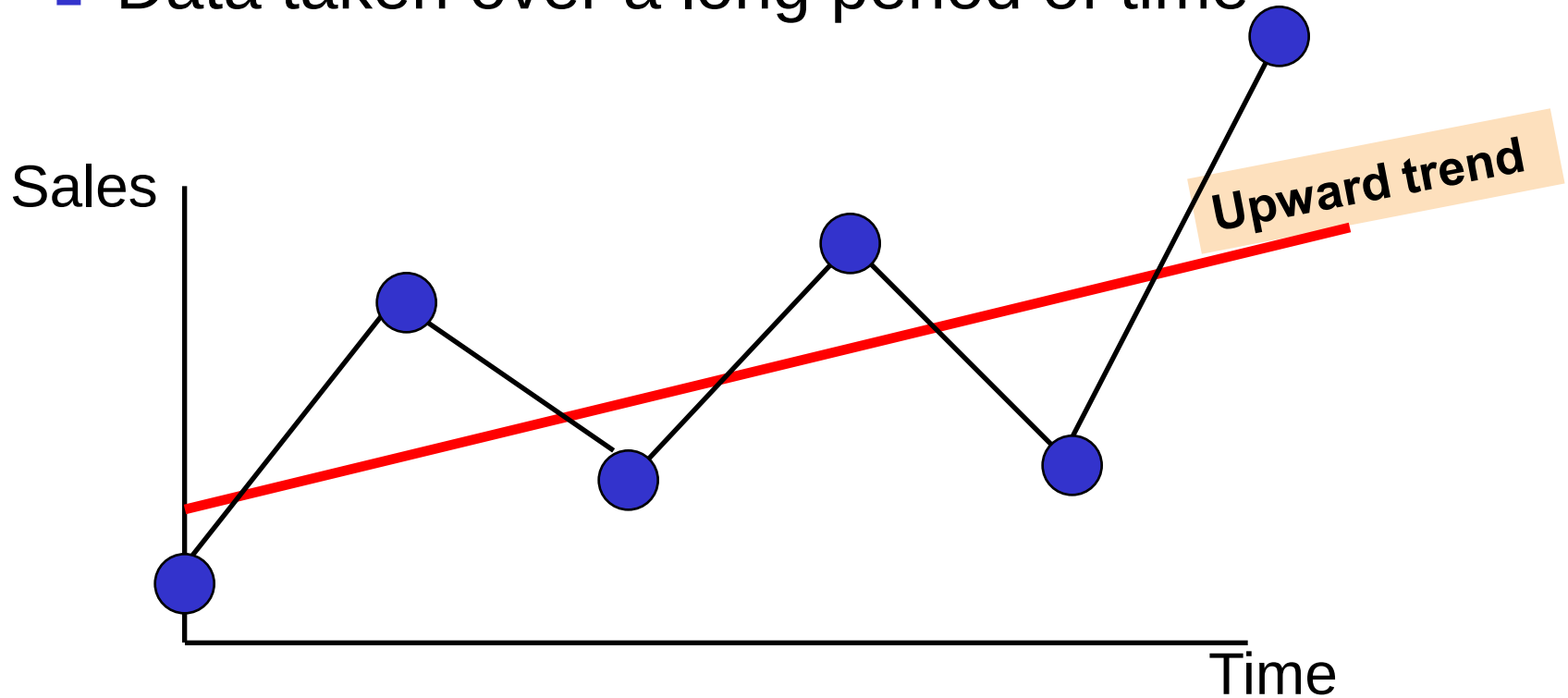


Time-Series Components



Trend Component

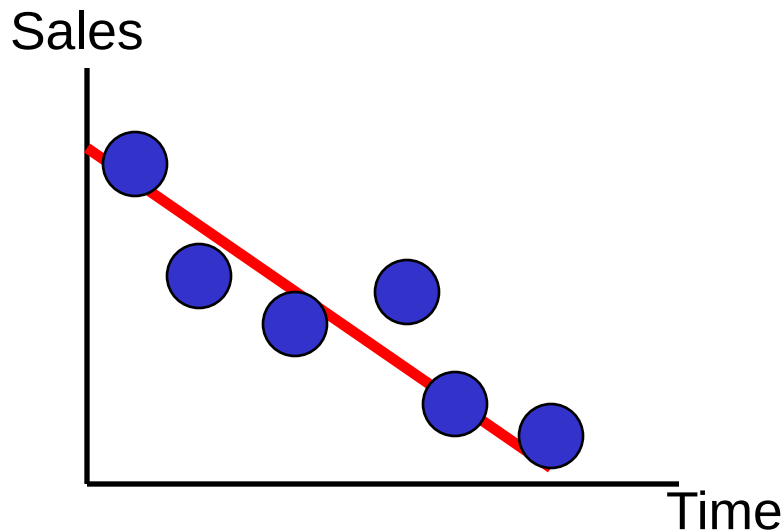
- Long-run increase or decrease over time (overall upward or downward movement)
- Data taken over a long period of time



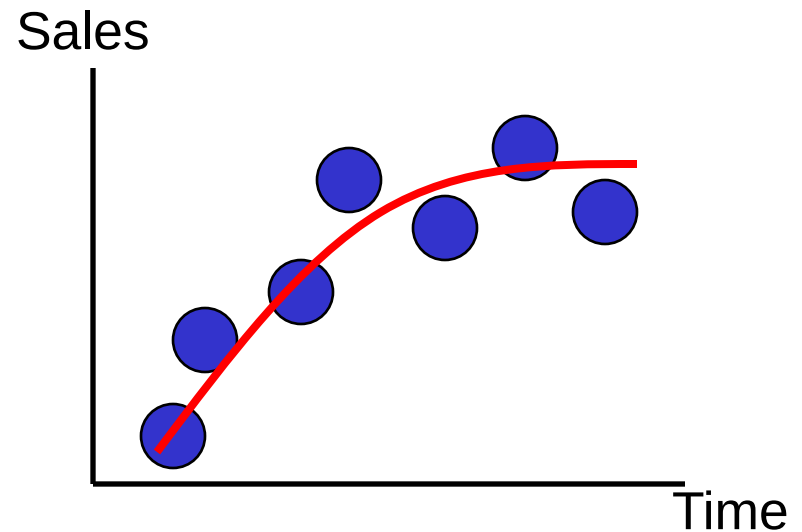
Trend Component

(continued)

- Trend can be upward or downward
- Trend can be linear or non-linear



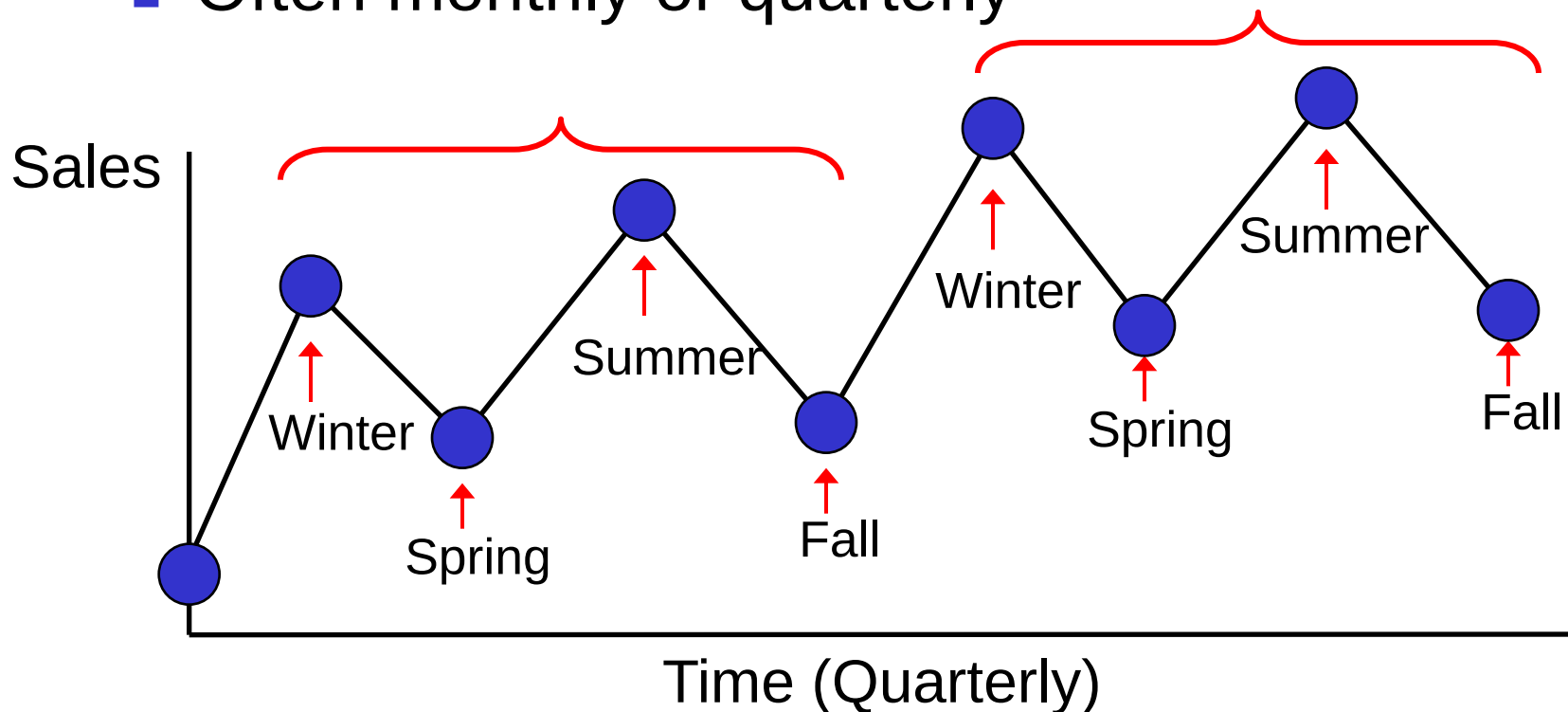
Downward linear trend



Upward nonlinear trend

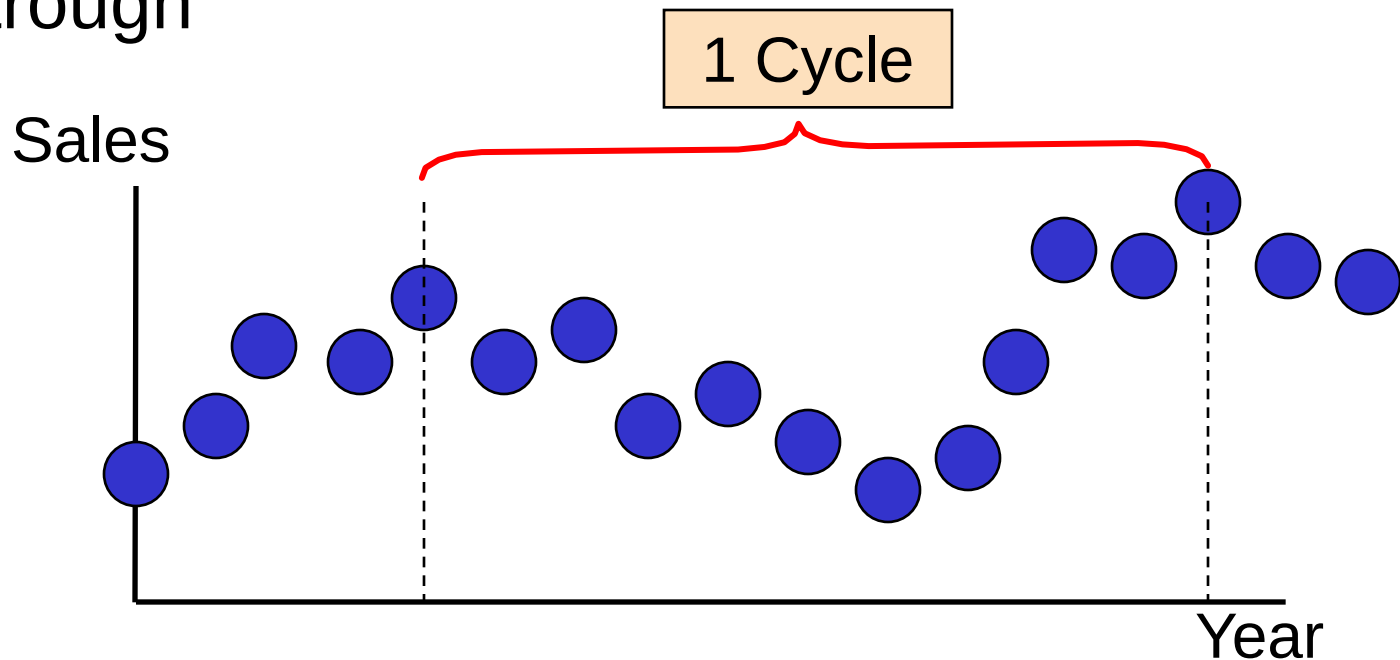
Seasonal Component

- Short-term regular wave-like patterns
- Observed within 1 year
- Often monthly or quarterly



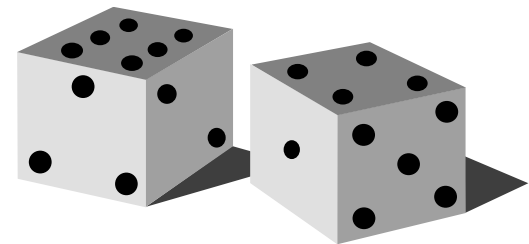
Cyclical Component

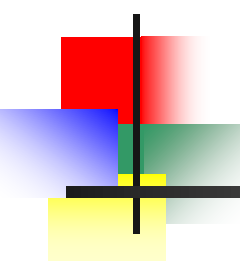
- Long-term wave-like patterns
- Regularly occur but may vary in length
- Often measured peak to peak or trough to trough



Irregular Component

- Unpredictable, random, “residual” fluctuations
- Due to random variations of
 - Nature
 - Accidents or unusual events
- “Noise” in the time series





Multiplicative Time-Series Model for Annual Data

- Used primarily for forecasting
- Observed value in time series is the product of components

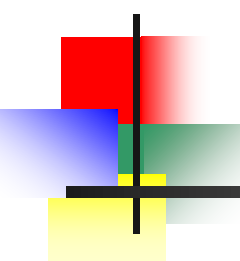
$$Y_i = T_i \times C_i \times I_i$$

where

T_i = Trend value at year i

C_i = Cyclical value at year i

I_i = Irregular (random) value at year i



Multiplicative Time-Series Model with a Seasonal Component

- Used primarily for forecasting
- Allows consideration of seasonal variation

$$Y_i = T_i \times S_i \times C_i \times I_i$$

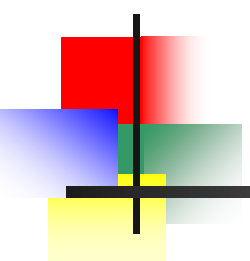
where

T_i = Trend value at time i

S_i = Seasonal value at time i

C_i = Cyclical value at time i

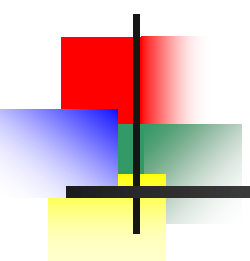
I_i = Irregular (random) value at time i



Smoothing the Annual Time Series

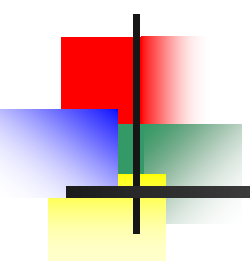
- Calculate moving averages to get an overall impression of the pattern of movement over time

Moving Average: averages of consecutive time series values for a chosen period of length L



Moving Averages

- Used for smoothing
- A series of arithmetic means over time
- Result dependent upon choice of L (length of period for computing means)
- Examples:
 - For a 5 year moving average, $L = 5$
 - For a 7 year moving average, $L = 7$
 - Etc.



Moving Averages

(continued)

- **Example:** Five-year moving average

- First average:

$$MA(5) = \frac{Y_1 + Y_2 + Y_3 + Y_4 + Y_5}{5}$$

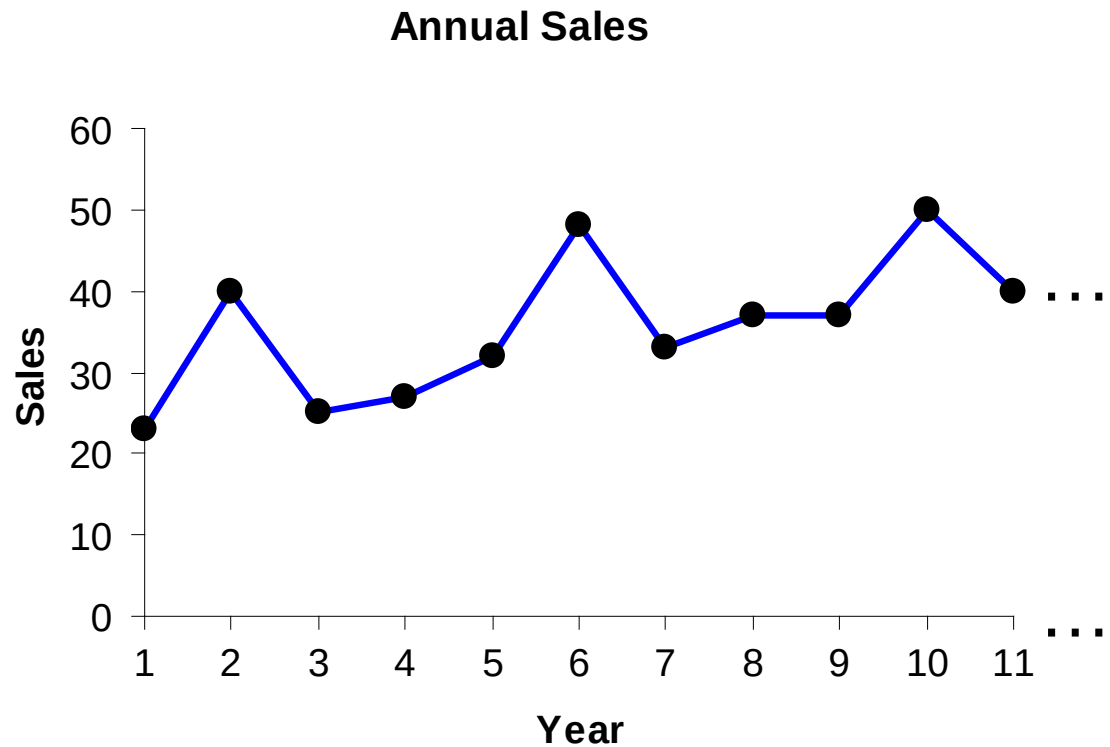
- Second average:

$$MA(5) = \frac{Y_2 + Y_3 + Y_4 + Y_5 + Y_6}{5}$$

- etc.

Example: Annual Data

Year	Sales
1	23
2	40
3	25
4	27
5	32
6	48
7	33
8	37
9	37
10	50
11	40
etc...	etc...



Calculating Moving Averages

Year	Sales
1	23
2	40
3	25
4	27
5	32
6	48
7	33
8	37
9	37
10	50
11	40

etc...

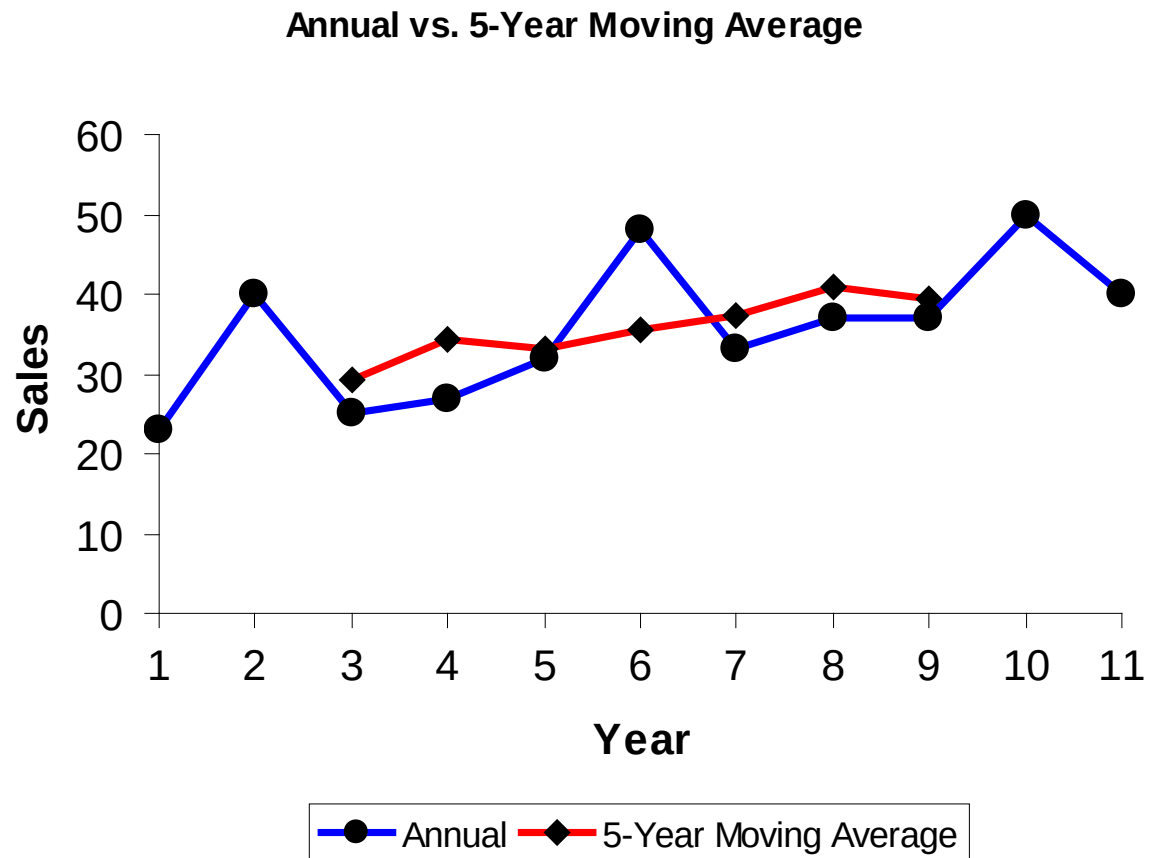
Average Year	5-Year Moving Average
3	29.4
4	34.4
5	33.0
6	35.4
7	37.4
8	41.0
9	39.4
...	...

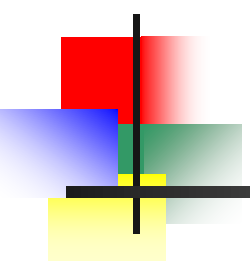
$$3 = \frac{1 + 2 + 3 + 4 + 5}{5}$$
$$29.4 = \frac{23 + 40 + 25 + 27 + 32}{5}$$

- Each moving average is for a consecutive block of 5 years

Annual vs. Moving Average

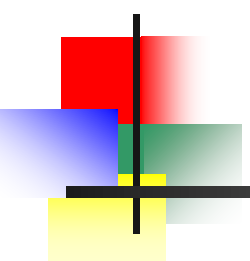
- The 5-year moving average smooths the data and shows the underlying trend





Exponential Smoothing

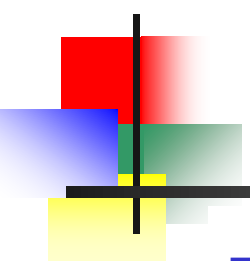
- A **weighted** moving average
 - Weights decline exponentially
 - Most recent observation weighted most
- Used for smoothing and short term forecasting (often one period into the future)



Exponential Smoothing

(continued)

- The weight (smoothing coefficient) is W
 - Subjectively chosen
 - Range from 0 to 1
 - Smaller W gives more smoothing, larger W gives less smoothing
- The weight is:
 - Close to 0 for smoothing out unwanted cyclical and irregular components
 - Close to 1 for forecasting



Exponential Smoothing Model

- Exponential smoothing model

$$E_1 = Y_1$$

$$E_i = WY_i + (1 - W)E_{i-1}$$

For $i = 2, 3, 4, \dots$

where:

E_i = exponentially smoothed value for period i

E_{i-1} = exponentially smoothed value already
computed for period $i - 1$

Y_i = observed value in period i

W = weight (smoothing coefficient), $0 < W < 1$

Exponential Smoothing Example

- Suppose we use weight $W = 0.2$

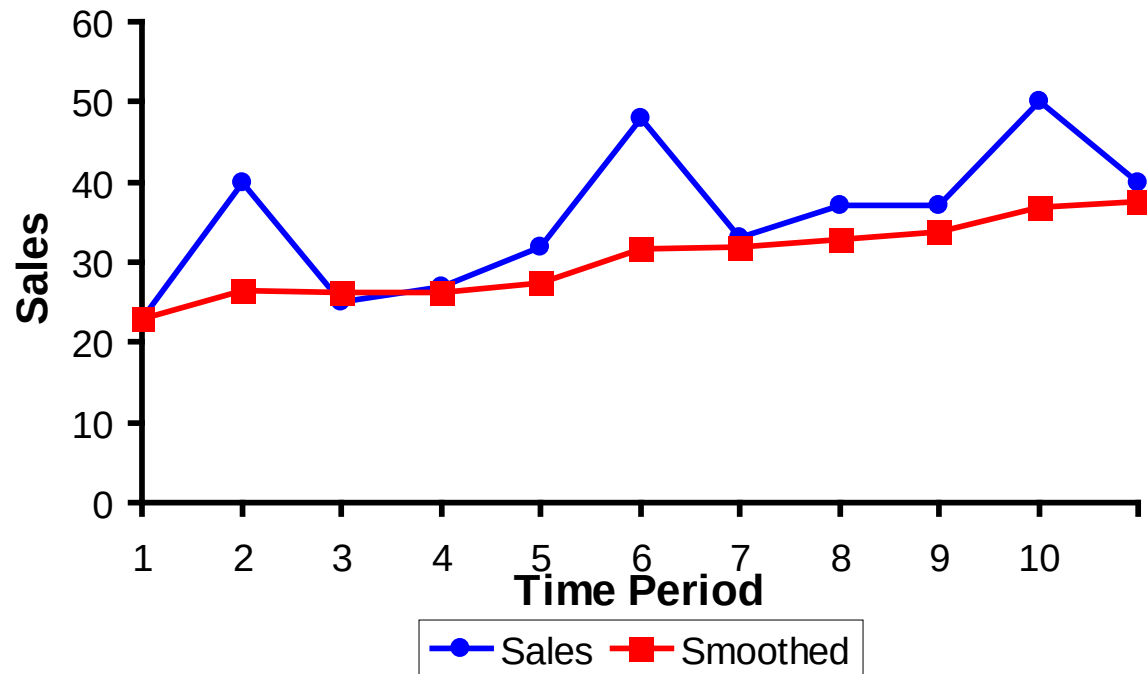
Time Period (i)	Sales (Y_i)	Forecast from prior period (E_{i-1})	Exponentially Smoothed Value for this period (E_i)
1	23	--	23
2	40	23	$(.2)(40) + (.8)(23) = 26.4$
3	25	26.4	$(.2)(25) + (.8)(26.4) = 26.12$
4	27	26.12	$(.2)(27) + (.8)(26.12) = 26.296$
5	32	26.296	$(.2)(32) + (.8)(26.296) = 27.437$
6	48	27.437	$(.2)(48) + (.8)(27.437) = 31.549$
7	33	31.549	$(.2)(33) + (.8)(31.549) = 31.840$
8	37	31.840	$(.2)(37) + (.8)(31.840) = 32.872$
9	37	32.872	$(.2)(37) + (.8)(32.872) = 33.697$
10	50	33.697	$(.2)(50) + (.8)(33.697) = 36.958$
etc.	etc.	etc.	etc.

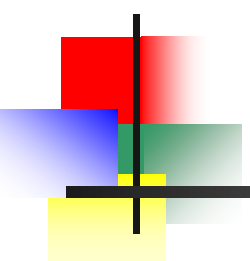
$E_1 = Y_1$
since no
prior
information
exists

$$E_i = WY_i + (1 - W)E_{i-1}$$

Sales vs. Smoothed Sales

- Fluctuations have been smoothed
- **NOTE:** the smoothed value in this case is generally a little low, since the trend is upward sloping and the weighting factor is only .2





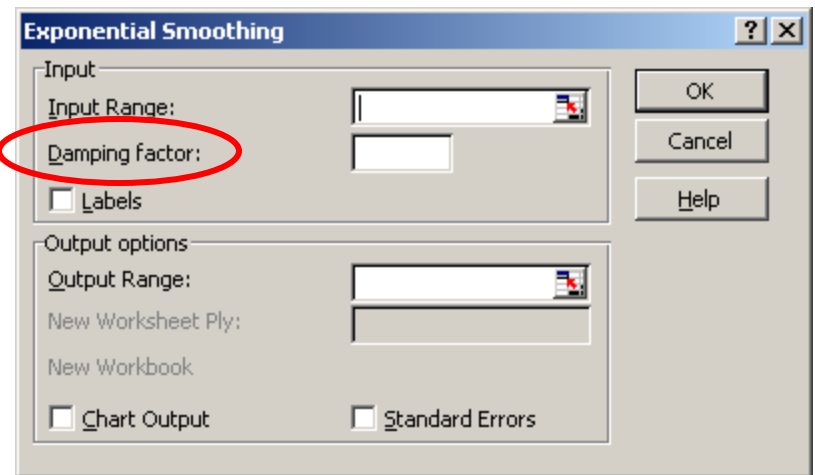
Forecasting Time Period $i + 1$

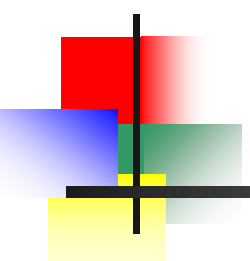
- The smoothed value in the current period (i) is used as the forecast value for next period ($i + 1$) :

$$\hat{Y}_{i+1} = E_i$$

Exponential Smoothing in Excel

- Use tools / data analysis / exponential smoothing
- The “damping factor” is $(1 - W)$





Trend-Based Forecasting

- Estimate a trend line using regression analysis

Year	Time Period (X)	Sales (Y)
1999	0	20
2000	1	40
2001	2	30
2002	3	50
2003	4	70
2004	5	65

- Use **time (X)** as the independent variable:

$$\hat{Y} = b_0 + b_1X$$

Trend-Based Forecasting

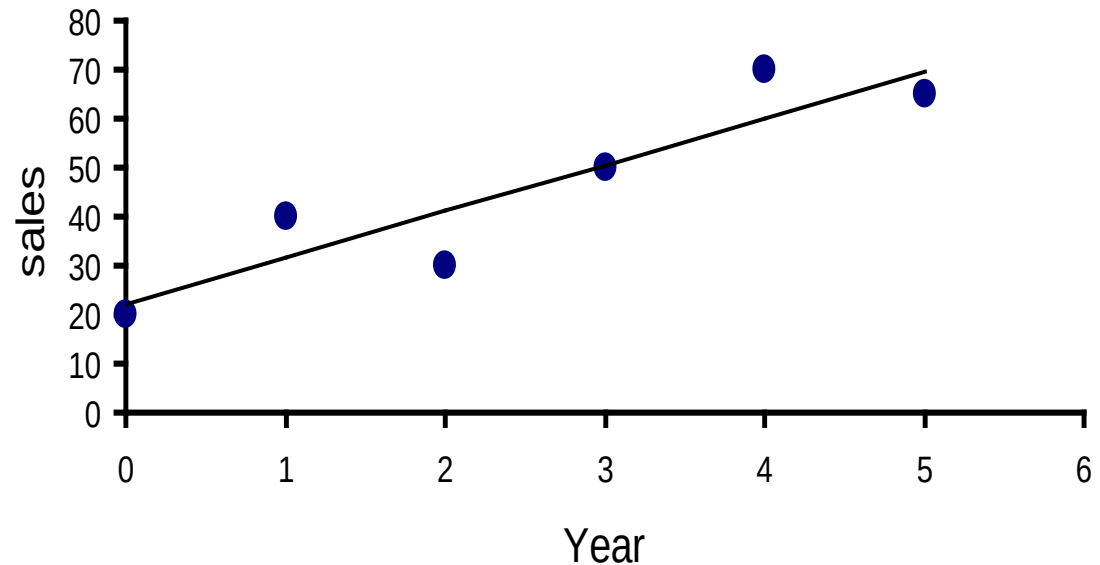
(continued)

Year	Time Period (X)	Sales (Y)
1999	0	20
2000	1	40
2001	2	30
2002	3	50
2003	4	70
2004	5	65

- The linear trend forecasting equation is:

$$\hat{Y}_i = 21.905 + 9.5714 X_i$$

Sales trend



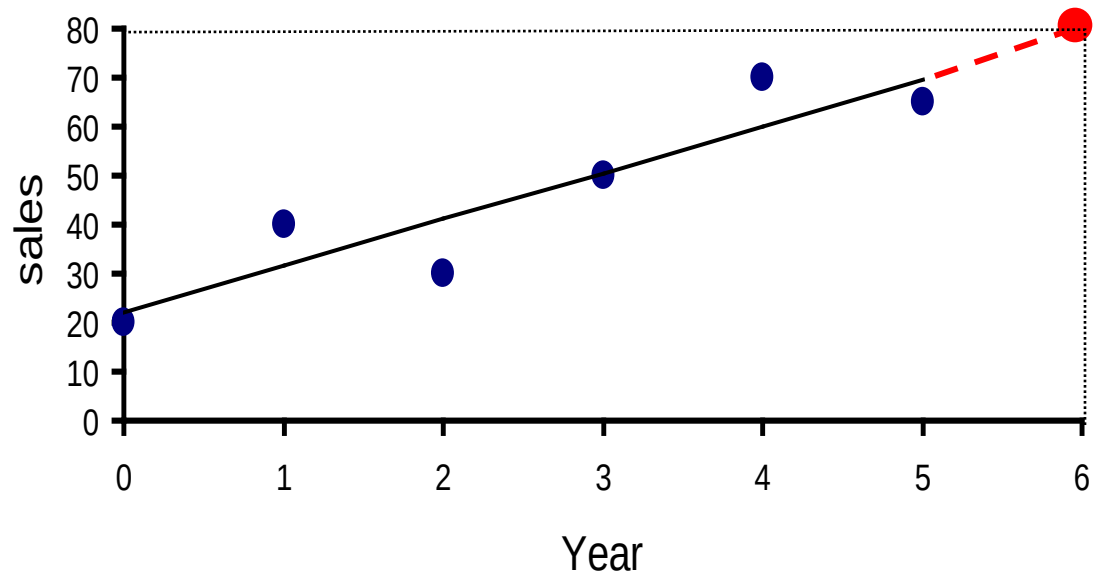
Trend-Based Forecasting

(continued)

- Forecast for time period 6:

$$\hat{Y} = 21.905 + 9.5714 (6)$$
$$= 79.33$$

Year	Time Period (X)	Sales (y)
1999	0	20
2000	1	40
2001	2	30
2002	3	50
2003	4	70
2004	5	65
2005	6	??



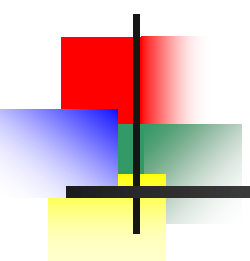


Nonlinear Trend Forecasting

- A nonlinear regression model can be used when the time series exhibits a nonlinear trend
- Quadratic form is one type of a nonlinear model:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$$

- Compare adj. r^2 and standard error to that of linear model to see if this is an improvement
- Can try other functional forms to get best fit



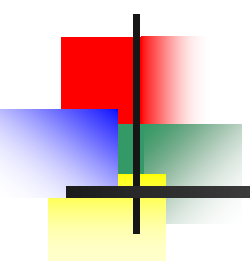
Exponential Trend Model

- Another nonlinear trend model:

$$Y_i = \beta_0 \beta_1^{X_i} \varepsilon_i$$

- Transform to linear form:

$$\log(Y_i) = \log(\beta_0) + X_i \log(\beta_1) + \log(\varepsilon_i)$$



Exponential Trend Model

(continued)

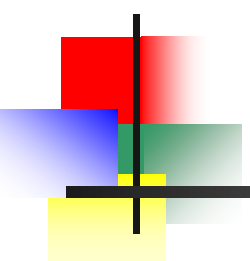
- Exponential trend forecasting equation:

$$\log(\hat{Y}_i) = b_0 + b_1 X_i$$

where b_0 = estimate of $\log(\beta_0)$
 b_1 = estimate of $\log(\beta_1)$

Interpretation:

$(\hat{\beta}_1 - 1) \times 100\%$ is the estimated annual compound growth rate (in %)



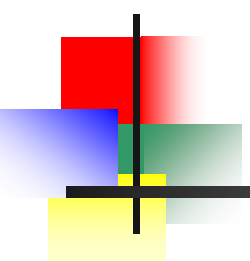
Model Selection Using Differences

- Use a linear trend model if the first differences are approximately constant

$$(Y_2 - Y_1) = (Y_3 - Y_2) = \cdots = (Y_n - Y_{n-1})$$

- Use a quadratic trend model if the second differences are approximately constant

$$\begin{aligned} [(Y_3 - Y_2) - (Y_2 - Y_1)] &= [(Y_4 - Y_3) - (Y_3 - Y_2)] \\ &= \cdots = [(Y_n - Y_{n-1}) - (Y_{n-1} - Y_{n-2})] \end{aligned}$$

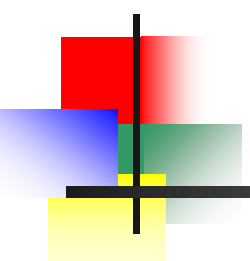


Model Selection Using Differences

(continued)

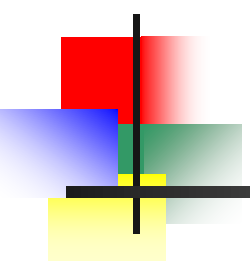
- Use an exponential trend model if the percentage differences are approximately constant

$$\frac{(Y_2 - Y_1)}{Y_1} \times 100\% = \frac{(Y_3 - Y_2)}{Y_2} \times 100\% = \dots = \frac{(Y_n - Y_{n-1})}{Y_{n-1}} \times 100\%$$



The Holt-Winters Method

- The Holt-Winters method extends exponential smoothing to include the future trend
- This method requires updated estimates of the time series value (the smoothed value E_i) and the trend value (T_i) in each period



The Holt-Winters Method

(continued)

- The Holt-Winters method:

$$\text{Level: } E_i = U(E_{i-1} + T_{i-1}) + (1 - U) Y_i$$

$$\text{Trend: } T_i = VT_{i-1} + (1 - V)(E_i - E_{i-1})$$

Where E_i = level of the smoothed series computed in period i

E_{i-1} = level of the smoothed series already computed in period $i - 1$

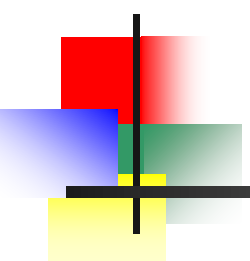
T_i = value of the trend component being computed in period i

T_{i-1} = value of the trend component already computed in period $i - 1$

Y_i = observed value in period i

U = subjectively assigned smoothing constant ($0 < U < 1$)

V = subjectively assigned smoothing constant ($0 < V < 1$)



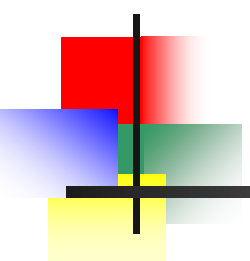
The Holt-Winters Method

(continued)

- To begin, define $E_2 = Y_2$ and $T_2 = Y_2 - Y_1$
- Choose smoothing constants U and V
- Compute E_i and T_i for all i years, $i = 3, 4, \dots, n$

Note:

- Smaller U values give more weight to more recent levels of the time series
- Smaller V values give more weight to the current trend and less weight to past trends in the series



Using the Holt-Winters Method for Forecasting

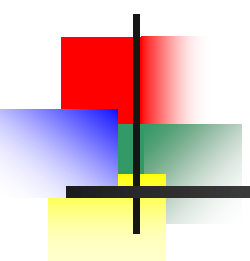
$$\hat{Y}_{n+j} = E_n + j(T_n)$$

Where $\hat{Y}_{n+j}$ = forecast value j years into the future

E_n = level of the smoothed series computed in the most recent time period n

T_n = value of the trend component computed in the most recent time period n

j = number of years into the future



Autoregressive Modeling

- Used for forecasting
- Takes advantage of autocorrelation
 - 1st order - correlation between consecutive values
 - 2nd order - correlation between values 2 periods apart
- p^{th} order Autoregressive model:

$$Y_i = A_0 + A_1 Y_{i-1} + A_2 Y_{i-2} + \cdots + A_p Y_{i-p} + \delta_i$$

Random
Error

Autoregressive Model: Example

The Office Concept Corp. has acquired a number of office units (in thousands of square feet) over the last eight years. Develop the **second order** Autoregressive model.

Year	Units
97	4
98	3
99	2
00	3
01	2
02	2
03	4
04	6



Autoregressive Model: Example Solution

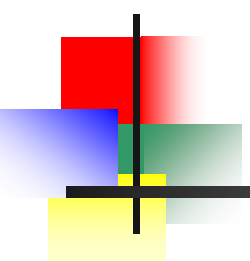
- Develop the 2nd order table
- Use Excel to estimate a regression model

Excel Output

	<i>Coefficients</i>
Intercept	3.5
X Variable 1	0.8125
X Variable 2	-0.9375

$$\hat{Y}_i = 3.5 + 0.8125Y_{i-1} - 0.9375Y_{i-2}$$

Year	Y_i	Y_{i-1}	Y_{i-2}
97	4	--	--
98	3	4	--
99	2	3	4
00	3	2	3
01	2	3	2
02	2	2	3
03	4	2	2
04	6	4	2

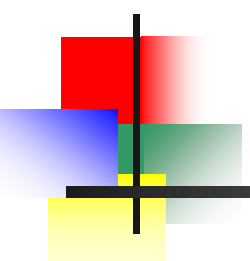


Autoregressive Model

Example: Forecasting

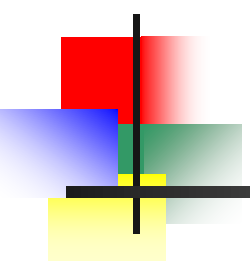
Use the second-order equation to forecast number of units for 2005:

$$\begin{aligned}\hat{Y}_i &= 3.5 + 0.8125Y_{i-1} - 0.9375Y_{i-2} \\ \hat{Y}_{2005} &= 3.5 + 0.8125(Y_{2004}) - 0.9375(Y_{2003}) \\ &= 3.5 + 0.8125(6) - 0.9375(4) \\ &= 4.625\end{aligned}$$



Autoregressive Modeling Steps

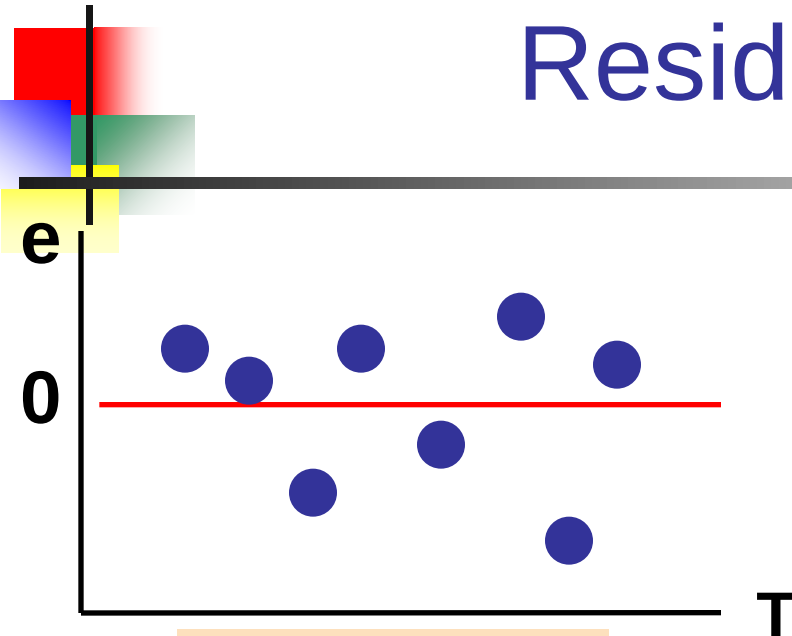
1. Choose p (note that $df = n - 2p - 1$)
2. Form a series of “lagged predictor” variables
 $Y_{i-1}, Y_{i-2}, \dots, Y_{i-p}$
3. Use Excel to run regression model using all p variables
4. Test significance of A_p
 - If null hypothesis rejected, this model is selected
 - If null hypothesis not rejected, decrease p by 1 and repeat



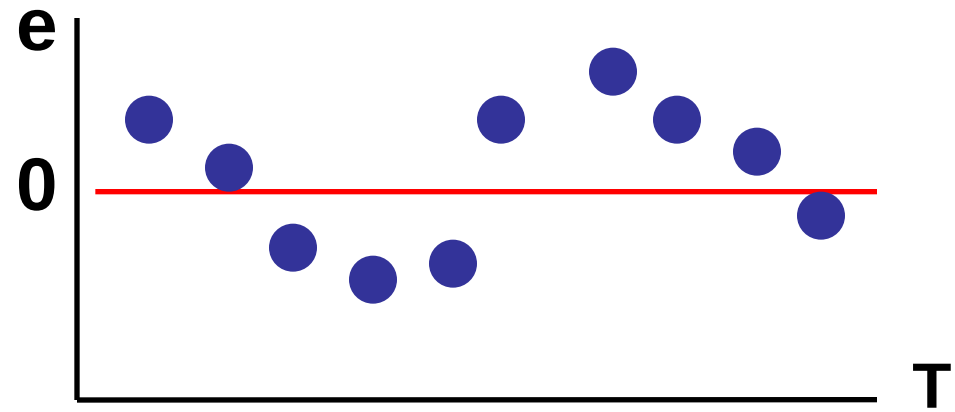
Choosing A Forecasting Model

- Perform a residual analysis
 - Look for pattern or direction
- Measure magnitude of residual error using squared differences
- Measure residual error using absolute differences
- Use simplest model
 - Principle of parsimony

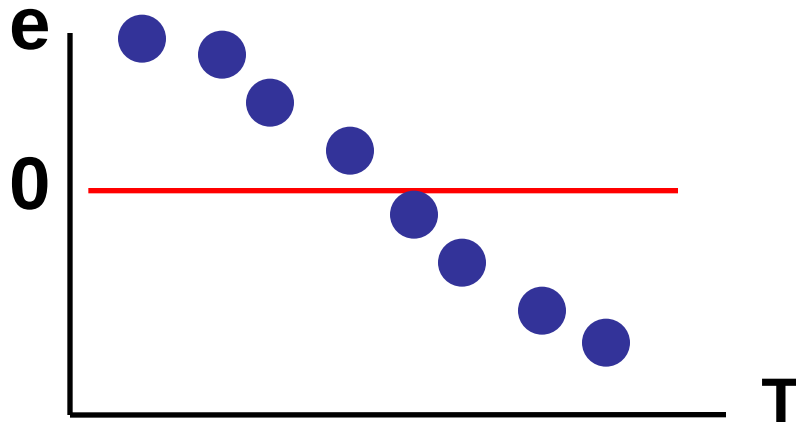
Residual Analysis



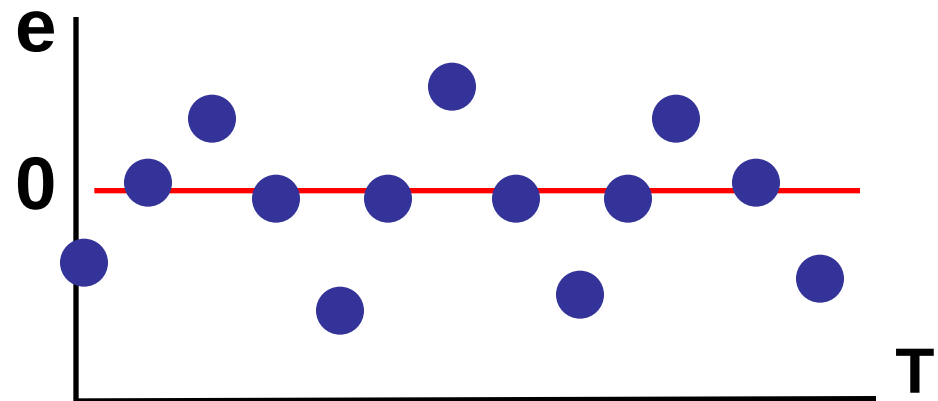
Random errors



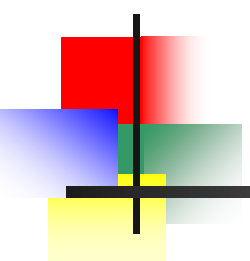
Cyclical effects not accounted for



Trend not accounted for



Seasonal effects not accounted for



Measuring Errors

- Choose the model that gives the smallest measuring errors

- Sum of squared errors (SSE)

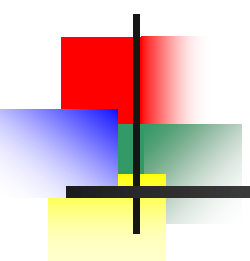
$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- Sensitive to outliers

- Mean Absolute Deviation (MAD)

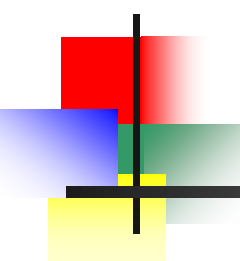
$$MAD = \frac{\sum_{i=1}^n |Y_i - \hat{Y}_i|}{n}$$

- Not sensitive to extreme observations



Principal of Parsimony

- Suppose two or more models provide a good fit for the data
- Select the simplest model
 - Simplest model types:
 - Least-squares linear
 - Least-squares quadratic
 - 1st order autoregressive
 - More complex types:
 - 2nd and 3rd order autoregressive
 - Least-squares exponential



Forecasting With Seasonal Data

- Recall the classical time series model with seasonal variation:

$$Y_i = T_i \times S_i \times C_i \times I_i$$

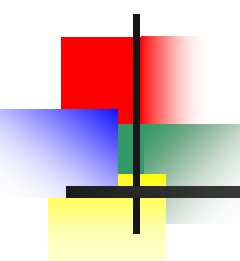
- Suppose the seasonality is quarterly
 - Define three new dummy variables for quarters:

$Q_1 = 1$ if first quarter, 0 otherwise

$Q_2 = 1$ if second quarter, 0 otherwise

$Q_3 = 1$ if third quarter, 0 otherwise

(Quarter 4 is the default if $Q_1 = Q_2 = Q_3 = 0$)



Exponential Model with Quarterly Data

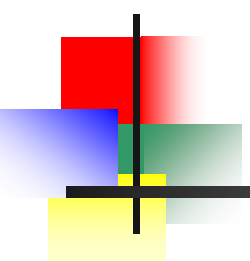
$$Y_i = \beta_0 \beta_1^{X_i} \beta_2^{Q_1} \beta_3^{Q_2} \beta_4^{Q_3} \varepsilon_i$$

$(\beta_1 - 1) \times 100\%$ is the quarterly compound growth rate

β_i provides the multiplier for the i^{th} quarter relative to the 4th quarter ($i = 2, 3, 4$)

■ Transform to linear form:

$$\begin{aligned} \log(Y_i) = & \log(\beta_0) + X_i \log(\beta_1) + Q_1 \log(\beta_2) \\ & + Q_2 \log(\beta_3) + Q_3 \log(\beta_4) + \log(\varepsilon_i) \end{aligned}$$



Estimating the Quarterly Model

- Exponential forecasting equation:

$$\log(\hat{Y}_i) = b_0 + b_1X_i + b_2Q_1 + b_3Q_2 + b_4Q_3$$

where b_0 = estimate of $\log(\beta_0)$, so $10^{b_0} = \hat{\beta}_0$
 b_1 = estimate of $\log(\beta_1)$, so $10^{b_1} = \hat{\beta}_1$
etc...

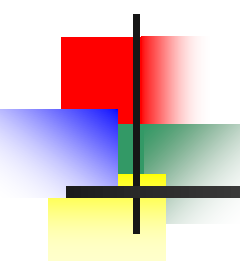
Interpretation:

$(\hat{\beta}_1 - 1) \times 100\%$ = estimated quarterly compound growth rate (in %)

$\hat{\beta}_2$ = estimated multiplier for first quarter relative to fourth quarter

$\hat{\beta}_3$ = estimated multiplier for second quarter rel. to fourth quarter

$\hat{\beta}_4$ = estimated multiplier for third quarter relative to fourth quarter



Quarterly Model Example

- Suppose the forecasting equation is:

$$\log(\hat{Y}_i) = 3.43 + .017X_i - .082Q_1 - .073Q_2 + .022Q_3$$

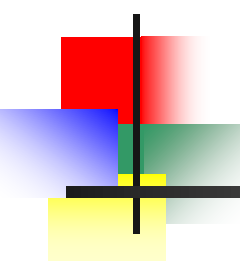
$$b_0 = 3.43, \text{ so } 10^{b_0} = \hat{\beta}_0 = 2691.53$$

$$b_1 = .017, \text{ so } 10^{b_1} = \hat{\beta}_1 = 1.040$$

$$b_2 = -.082, \text{ so } 10^{b_2} = \hat{\beta}_2 = 0.827$$

$$b_3 = -.073, \text{ so } 10^{b_3} = \hat{\beta}_3 = 0.845$$

$$b_4 = .022, \text{ so } 10^{b_4} = \hat{\beta}_4 = 1.052$$



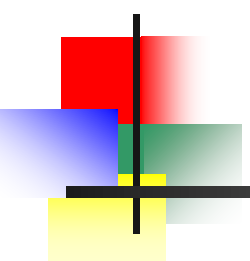
Quarterly Model Example

(continued)

Value:

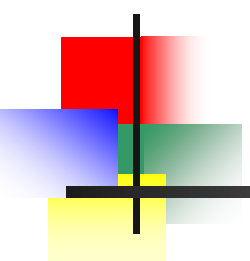
Interpretation:

$\hat{\beta}_0 = 2691.53$	Unadjusted trend value for first quarter of first year
$\hat{\beta}_1 = 1.040$	4.0% = estimated quarterly compound growth rate
$\hat{\beta}_2 = 0.827$	Ave. sales in Q_2 are 82.7% of average 4 th quarter sales, after adjusting for the 4% quarterly growth rate
$\hat{\beta}_3 = 0.845$	Ave. sales in Q_3 are 84.5% of average 4 th quarter sales, after adjusting for the 4% quarterly growth rate
$\hat{\beta}_4 = 1.052$	Ave. sales in Q_4 are 105.2% of average 4 th quarter sales, after adjusting for the 4% quarterly growth rate



Index Numbers

- Index numbers allow **relative** comparisons over time
- Index numbers are reported relative to a **Base Period Index**
- Base period index = 100 by definition
- Used for an individual item or measurement



Simple Price Index

- Simple Price Index:

$$I_i = \frac{P_i}{P_{\text{base}}} \times 100$$

where

I_i = index number for year i

P_i = price for year i

P_{base} = price for the base year

Index Numbers: Example

- Airplane ticket prices from 1995 to 2003:

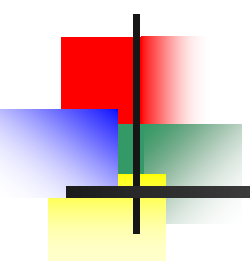
Year	Price	Index (base year = 2000)
1995	272	85.0
1996	288	90.0
1997	295	92.2
1998	311	97.2
1999	322	100.6
2000	320	100.0
2001	348	108.8
2002	366	114.4
2003	384	120.0

$$I_{1996} = \frac{P_{1996}}{P_{2000}} \times 100 = \frac{288}{320}(100) = 90$$

Base Year:

$$I_{2000} = \frac{P_{2000}}{P_{2000}} \times 100 = \frac{320}{320}(100) = 100$$

$$I_{2003} = \frac{P_{2003}}{P_{2000}} \times 100 = \frac{384}{320}(100) = 120$$



Index Numbers: Interpretation

$$I_{1996} = \frac{P_{1996}}{P_{2000}} \times 100 = \frac{288}{320}(100) = 90$$

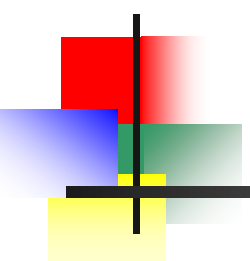
- Prices in 1996 were 90% of base year prices

$$I_{2000} = \frac{P_{2000}}{P_{2000}} \times 100 = \frac{320}{320}(100) = 100$$

- Prices in 2000 were 100% of base year prices (by definition, since 2000 is the base year)

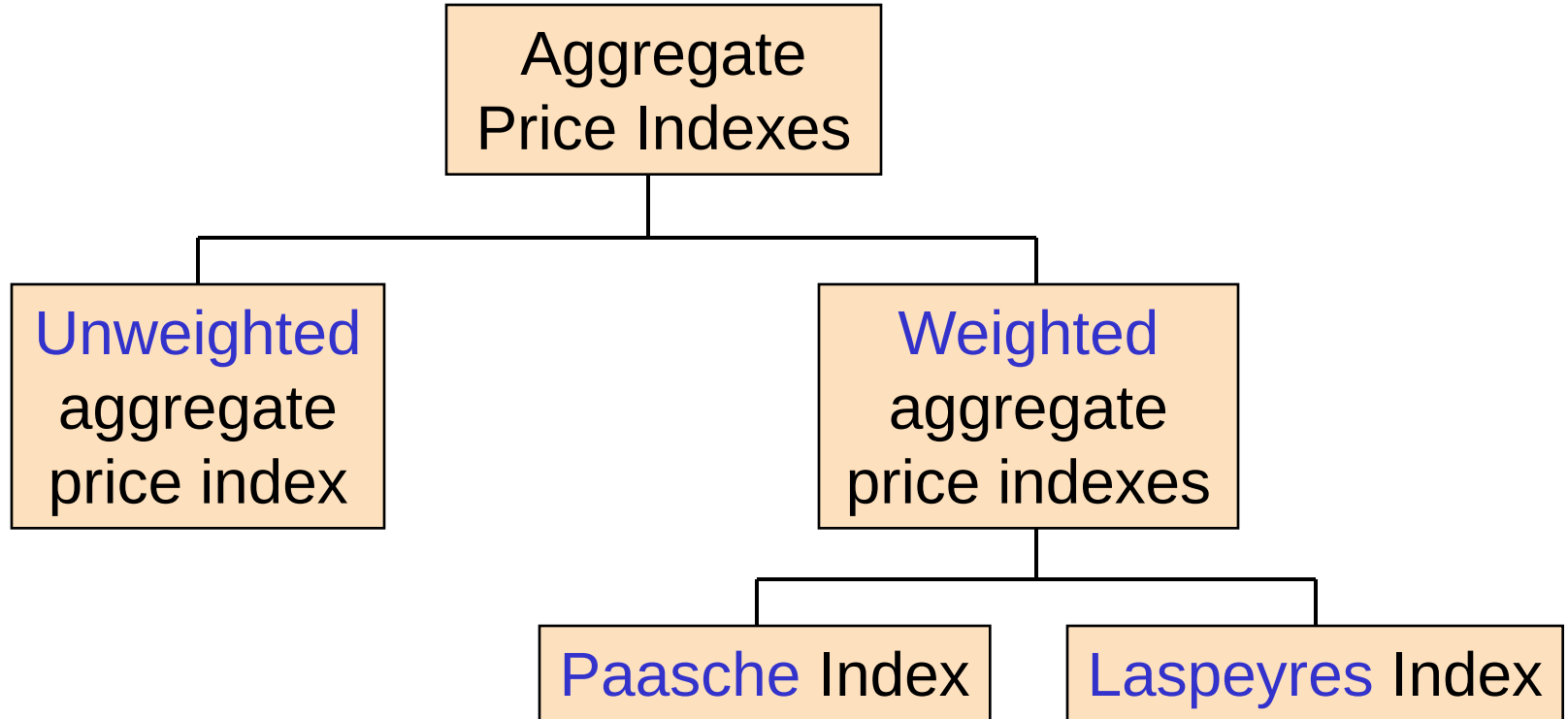
$$I_{2003} = \frac{P_{2003}}{P_{2000}} \times 100 = \frac{384}{320}(100) = 120$$

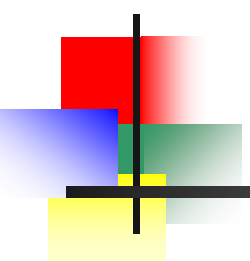
- Prices in 2003 were 120% of base year prices



Aggregate Price Indexes

- An **aggregate index** is used to measure the rate of change from a base period for a **group of items**





Unweighted Aggregate Price Index

- Unweighted aggregate price index formula:

$$I_U^{(t)} = \frac{\sum_{i=1}^n P_i^{(t)}}{\sum_{i=1}^n P_i^{(0)}} \times 100$$

i = item

t = time period

n = total number of items

$I_U^{(t)}$ = unweighted price index at time t

$\sum_{i=1}^n P_i^{(t)}$ = sum of the prices for the group of items at time t

$\sum_{i=1}^n P_i^{(0)}$ = sum of the prices for the group of items in time period 0

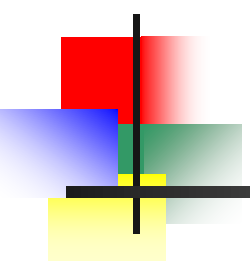
Unweighted Aggregate Price Index: Example

	Automobile Expenses: Monthly Amounts (\$):				
Year	Lease payment	Fuel	Repair	Total	Index (2001=100)
2001	260	45	40	345	100.0
2002	280	60	40	380	110.1
2003	305	55	45	405	117.4
2004	310	50	50	410	118.8



$$I_{2004} = \frac{\sum P_{2004}}{\sum P_{2001}} \times 100 = \frac{410}{345} (100) = 118.8$$

- Unweighted total expenses were 18.8% higher in 2004 than in 2001



Weighted Aggregate Price Indexes

■ Laspeyres index

$$I_L^{(t)} = \frac{\sum_{i=1}^n P_i^{(t)} Q_i^{(0)}}{\sum_{i=1}^n P_i^{(0)} Q_i^{(0)}} \times 100$$

$Q_i^{(0)}$: weights based on
period 0 quantities

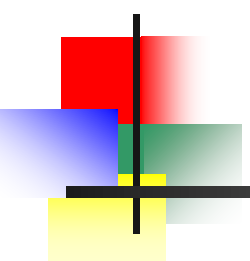
■ Paasche index

$$I_P^{(t)} = \frac{\sum_{i=1}^n P_i^{(t)} Q_i^{(t)}}{\sum_{i=1}^n P_i^{(0)} Q_i^{(t)}} \times 100$$

$Q_i^{(t)}$: weights based on current
period quantities

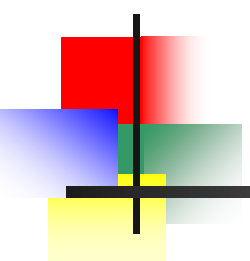
$P_i^{(t)}$ = price in time period t

$P_i^{(0)}$ = price in period 0



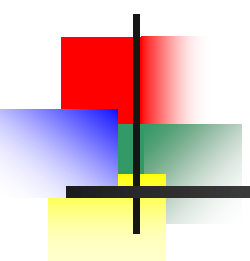
Common Price Indexes

- Consumer Price Index (CPI)
- Producer Price Index (PPI)
- Stock Market Indexes
 - Dow Jones Industrial Average
 - S&P 500 Index
 - NASDAQ Index



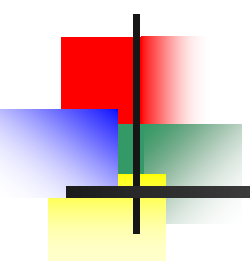
Pitfalls in Time-Series Analysis

- Assuming the mechanism that governs the time series behavior in the past will still hold in the future
- Using mechanical extrapolation of the trend to forecast the future without considering personal judgments, business experiences, changing technologies, and habits, etc.



Chapter Summary

- Discussed the importance of forecasting
- Addressed component factors of the time-series model
- Performed smoothing of data series
 - Moving averages
 - Exponential smoothing
- Described least square trend fitting and forecasting
 - Linear, quadratic and exponential models



Chapter Summary

(continued)

- Described the Holt-Winters method
- Addressed autoregressive models
- Described procedure for choosing appropriate models
- Addressed time series forecasting of monthly or quarterly data (use of dummy variables)
- Discussed pitfalls concerning time-series analysis